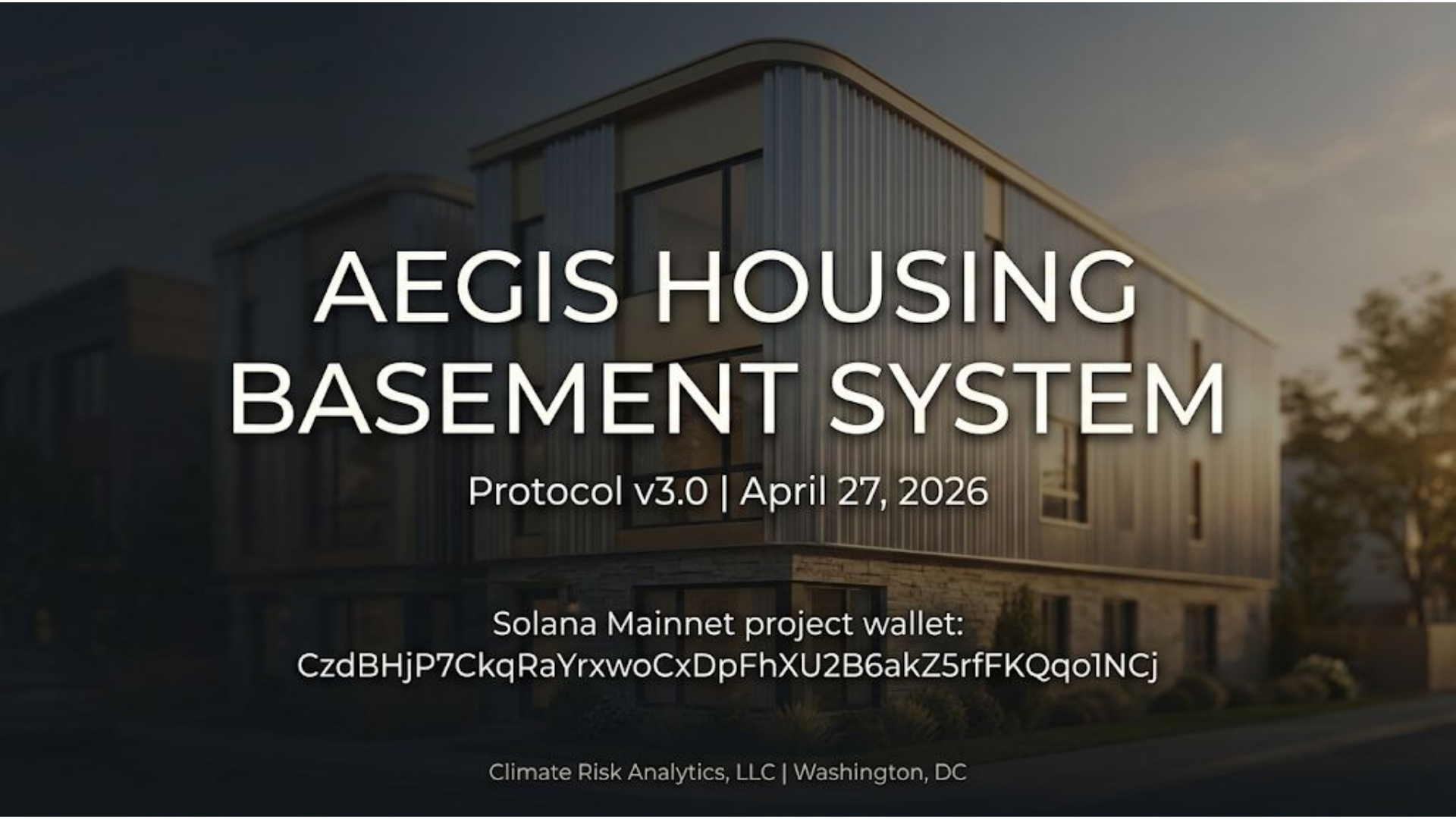




AEGIS HOUSING BASEMENT SYSTEM

Protocol v3.0 | April 27, 2026

Solana Mainnet project wallet:
`CzdBHjP7CkqRaYrxwoCxDpFhXU2B6akZ5rfFKQqo1NCj`

Climate Risk Analytics, LLC | Washington, DC

Who I Am



Martin Poveda Amarfil
Climate Risk Analytics, LLC



**Maritime &
Aerospace**

Defense

**Energy
Infrastructure**

**Climate-Resilient
Housing**

I am Martin Poveda Amarfil, founder of Climate Risk Analytics, LLC (CRA) — an AI-mature engineering design firm specializing in AI-integrated engineering for maritime and aerospace, defense, energy infrastructure, and climate-resilient housing systems.

Core Skills & Technical Standing (Q1 2026)



Extreme Outlier,
z-score > +3.0 σ



14–20x

Productivity Multiplier

Compressing weeks of complex
engineering into hours



Exterior Rear Elevation



AEGIS HOUSING BASEMENT SYSTEM

Why This Aegis House & Basement Matter – Energy Reduction & Tokenization

The Aegis Housing System: Climate-resilient, ultra-low-energy, equity-building home.

ENERGY REDUCTION



Zero-Utility HVAC
(50-70%+ Energy Savings)

200-400+ ft
Depth

~55°F

Constant Earth Temp

High-density
polyethylene
U-loop

Water-to-Air/
Water-to-Water
Heat Pump

Completed construction

Geothermal Install

Geothermal Install

Foundation

Blockchain



On-Chain Tokenization

(SHA-256 Proof-of-Work, RWA Assets)

**Pre-Paid Energy
Infrastructure & Future
Battery Tie-in**



ICF Walls

(6-8" Reinforced Concrete,
R-24+, Disaster Resilient)

EQUITY BUILDING

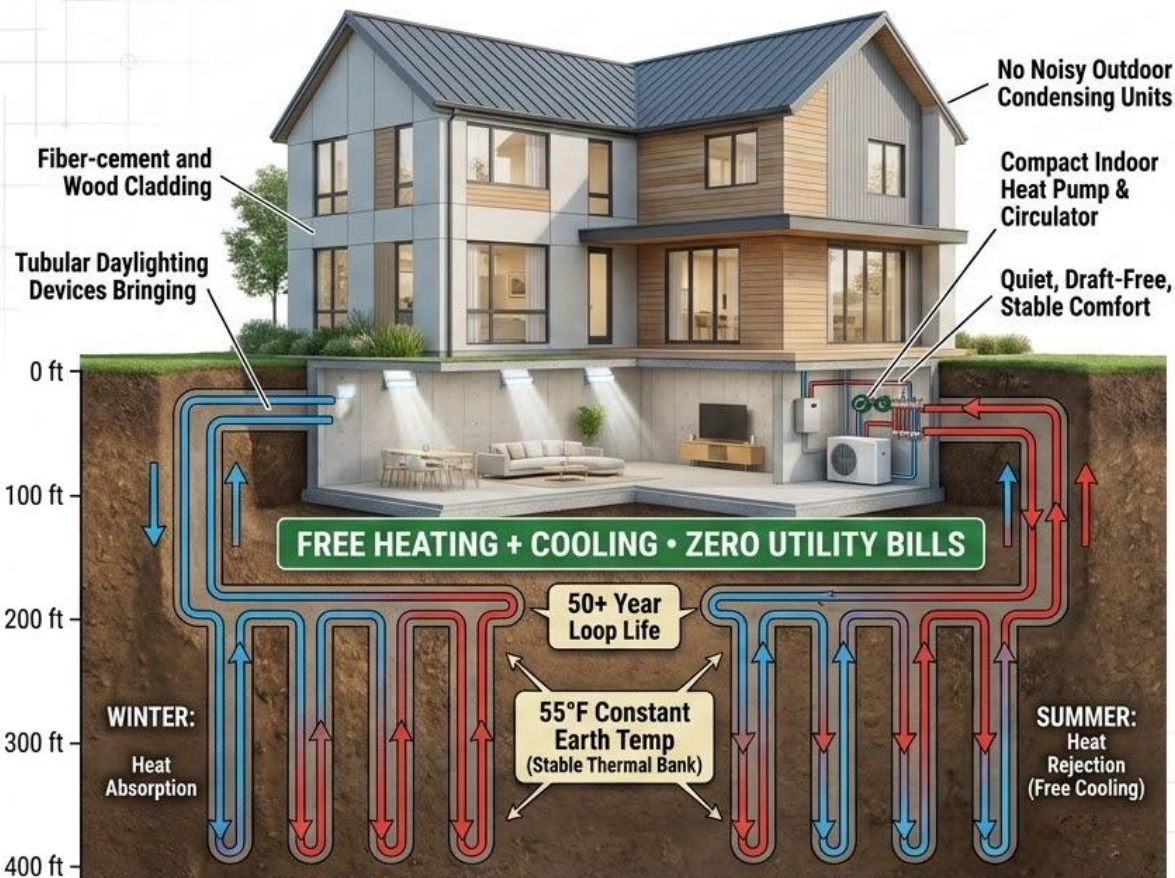


**Zero Water
Intrusion System**

This basement is the proof layer where science, resilience, energy independence, and decentralized validation converge.

ZERO UTILITY BILLS – GEOTHERMAL ENGINEERING (MID-ATLANTIC SPECS)

Geothermal Passive Heating & Free Cooling – The AEGIS HOUSING SYSTEM



AEGIS HOUSING SYSTEM: TECHNICAL SPECIFICATIONS		
CATEGORY	SPECIFICATION	VALUE / DESCRIPTION
SYSTEM CORE	Basement Depth	10 ft (3 m)
	Embedded Loops	~1,400 ft of 1" HDPE in concrete walls & slab
PERFORMANCE	Heating/Cooling Capacity	55,000–100,000 BTU/hr
	Passive Mode Power	<100 W (circulator only)
	Supplemental Mode	Low-power GSHP (Target COP 4.2)
	Energy Savings	60–70% vs. conventional HVAC
KEY BENEFITS	HVAC Utility Bills	ZERO
	Outdoor Equipment	NONE (No noise, no visual clutter)
	Loop Lifespan	50+ Years (DOE/EPA Validated)
CONSTRUCTION	Core Build Time	~40 operating hours (mini-excavator)
	Flood Resistance	Multi-layer drainage, waterproofing, R-15+
	Compliance	2021 VA USBC / IECC / IRC

Exterior Front Elevation



AEGIS HOUSING BASEMENT SYSTEM

Living Room Interior



AEGIS HOUSING BASEMENT SYSTEM

TITLE: AEGIS HOUSING SYSTEM BASEMENT – Passive Geothermal Thermal Core – The Heart of the House

Hybrid Drawing 1: Exploded Axonometric Basement Section

One-Piece Inverted-U Prefabricated Shell
(Marine-Grade Corrugated Stainless Steel)
Solar-Ready Sections
Zero Programmed Maintenance

Zoned Circadian LED Lighting System
with daylight harvesting & occupancy sensors, mood, productivity, sleep

Warm Wood Brown
Cedar/Accoya cladding - cabinets
Heamr-indoor/outdoor flow

Natural Stone Gray
+ Stone veneer walls
+ sland/Intesatcr wals
Human-Centric Coral
-Human-Centric Coral:
Indoorr/outdoor flow

Engineered Drainage Swale/Bioswale
Infinite passive energy storage heat exchanger, earth-stabilized 50-55 °F temperatures

Exterior Waterproofing System
Multi-layer membrane + dimple mat + R-15+ exterior insulation, Flood-resistant below-grade protection, zero-maintenance

One-Piece Inverted-U Prefabricated Shell
Roof + Long Angled Side/Party Walls (Integrated)
Seamless **Monolithic** module
Zero Programmed Maintenance

Tubular Daylighting Devices (6-10 units) optimized for human well-being, abundant natural light to basement & interiors

Personal House Equity Account

All 70-80 % Energy Savings and Centralized 45-House Mini-Grid Efficiencies Flow Directly into your personal house equity account (centralized mini-grid icon with 45 units serving large solar array/ central battery bank)

High-Mass Reinforced Concrete (Thermal Core) 80-95 cu yd
Basement structural foundation + infinite thermal bank

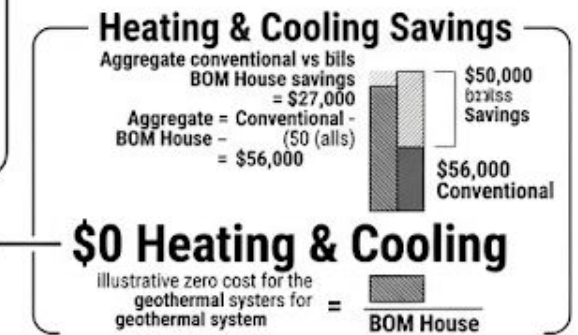
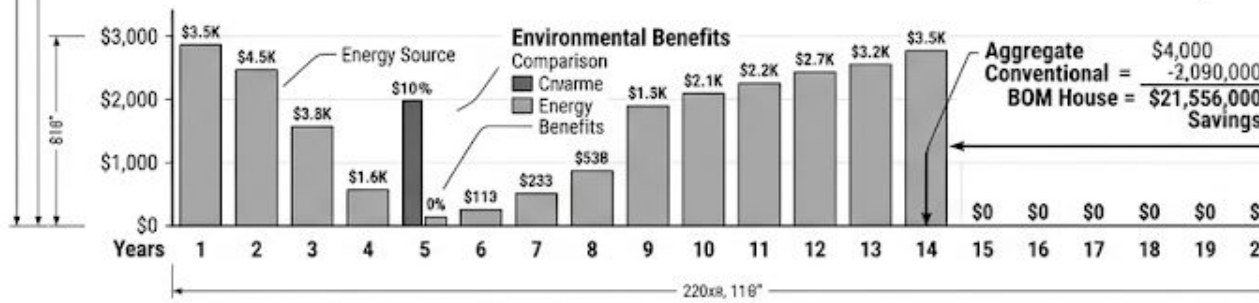
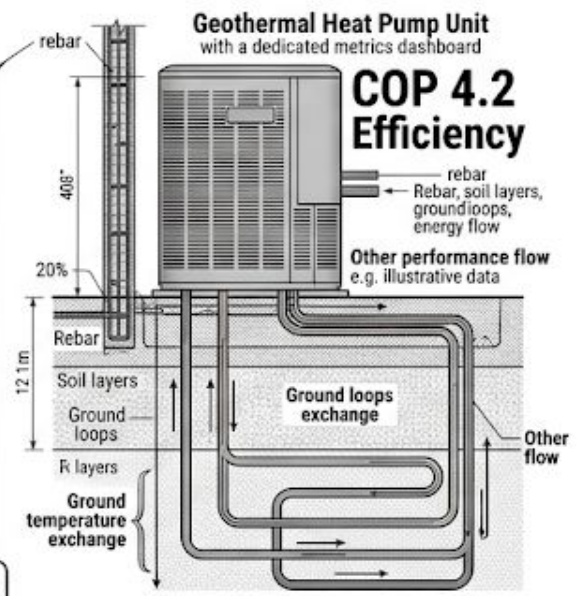
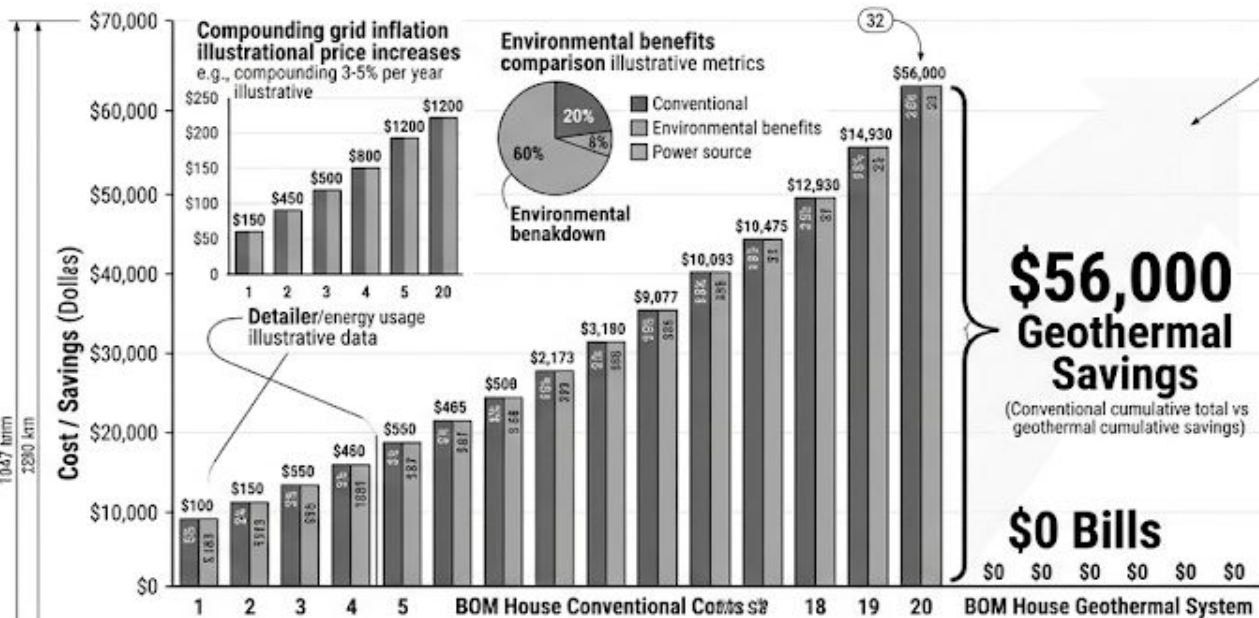
Embedded Geothermal HDPE Loops, Total, 1,400 linear ft of 1" SDR-11 HDPE pipe

Embedded Geothermal HDPE Loops, Manifold, <100 W circulator (passive mode), and optional Low-power GSHP (target COP 4.2) fully integrated

Bright usable living space
Bright usable living space, structural foundation, energy plant, zero maintenance

Hybrid Drawing 1: Exploded Axonometric Basement Section of AEGIS Housing System

20-Year Heating & Cooling Savings



NAVY-GRADE STRUCTURAL ENGINEERING

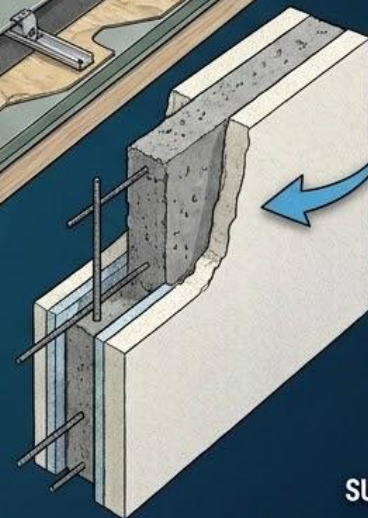
Structural Efficiencies & 50-Year Guarantee

Engineered Features:

- Insulated Concrete Forms (ICF) foundation & walls with 6-8" reinforced concrete core (R-22 to R-24+ whole-wall assembly)
- Heavy-gauge (24-gauge) standing-seam metal roofs with Galvalume® AZ55 coating – up to 50-year substrate warranty
- Modular prefab components for rapid, precise assembly and cost control
- Disaster resilience: 200-250 mph wind resistance, 4-hour fire rating, superior impact protection



Standing-seam
Galvalume panels
(24-gauge AZ55)



ICF wall layers
(foam + 6-8"
concrete core)



200-250 MPH
WIND RESISTANCE



4-HOUR
FIRE RATING



SUPERIOR IMPACT
PROTECTION

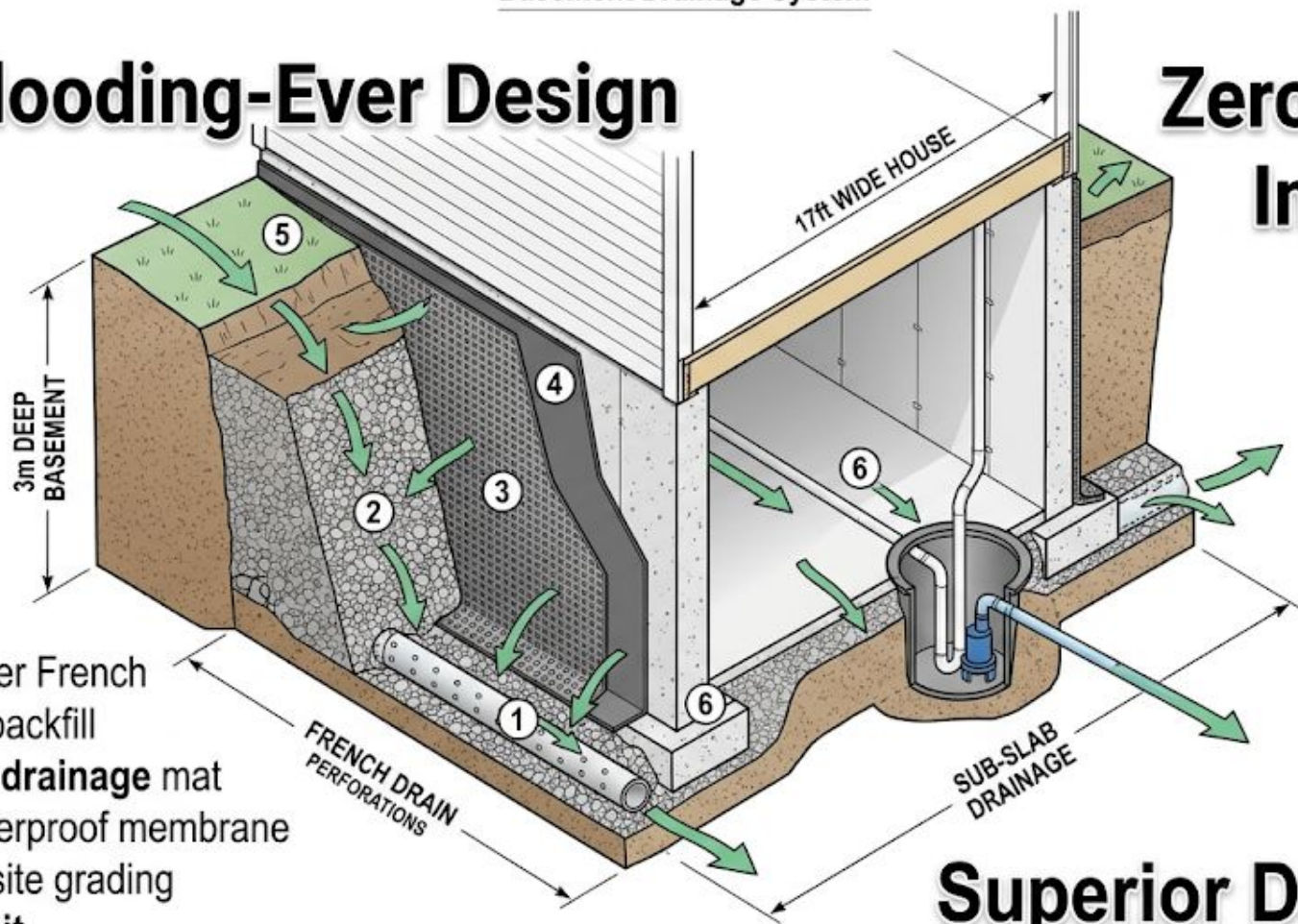


SUPERIOR WITHICAL
PROTECTION

Result: 50-year structural & roofing performance guarantee – far exceeding code minimums.

No-Flooding-Ever Design

Zero Water Ingress



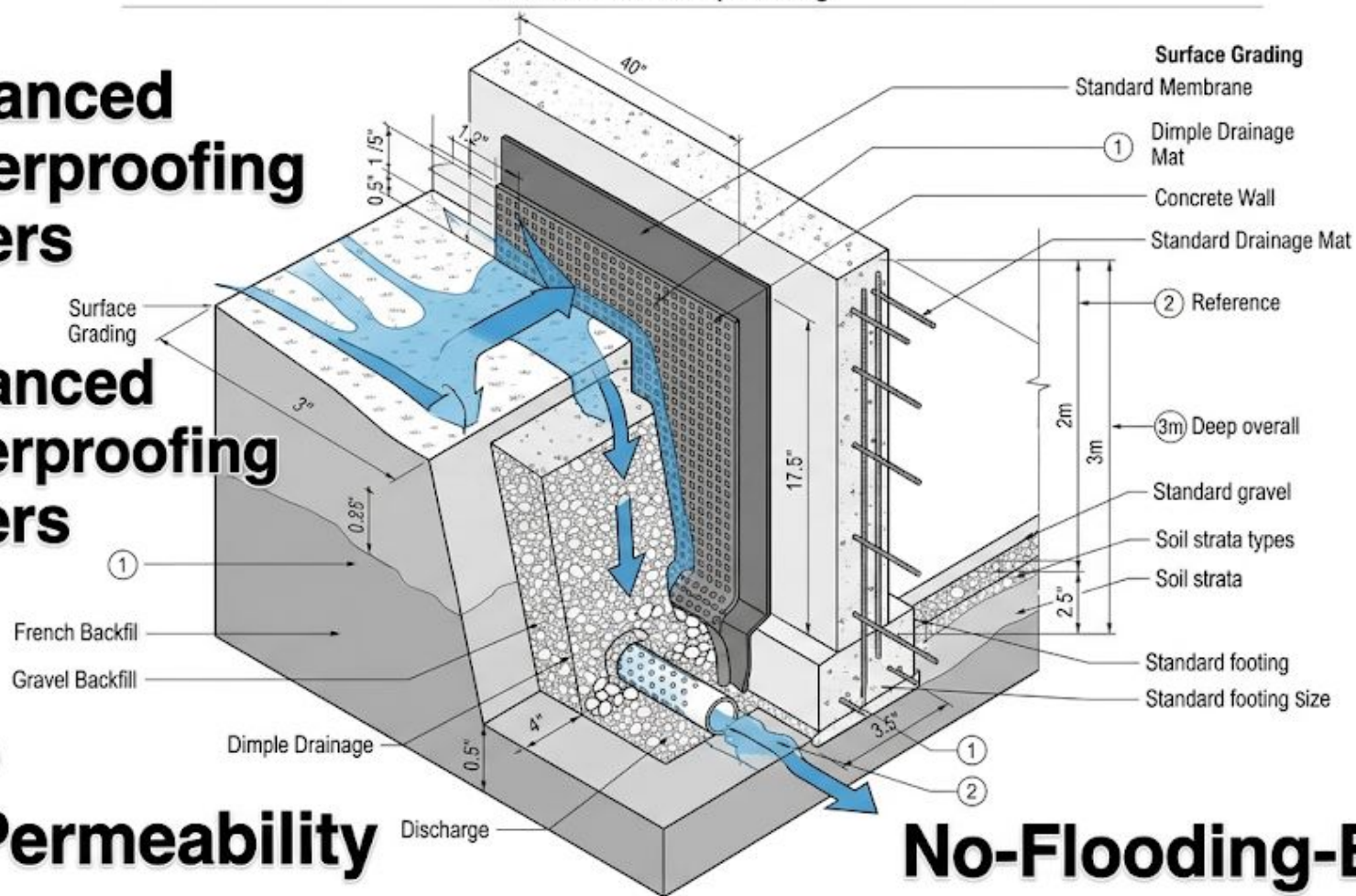
Superior Drainage

Basement Waterproofing

**Advanced
Waterproofing
Layers**

**Advanced
Waterproofing
Layers**

**Zero
No-Permeability**

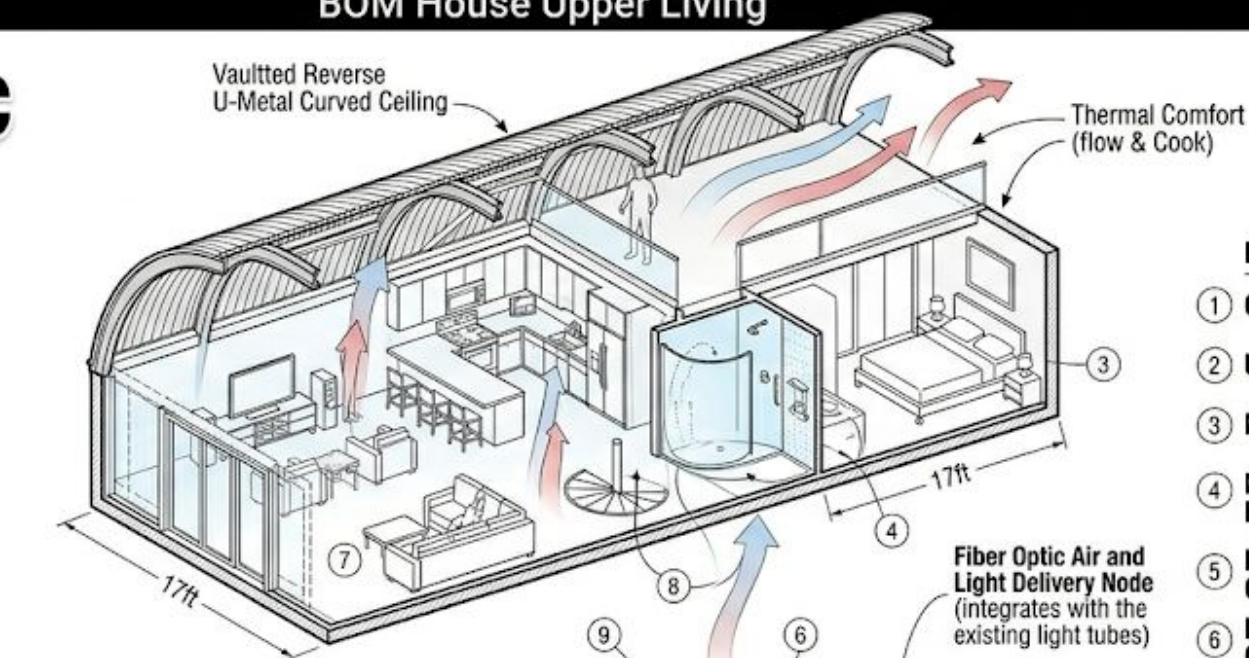


No-Flooding-Ever

Aesthetic Human Living

Thermal Comfort

House Stays Warm & Cool



LEGEND

- ① Open Great Room Area
- ② U-shaped Kitchen
- ③ Master Bedroom
- ④ Master Bathroom with Rounded Epoxy Shower
- ⑤ Loft overlooking Great Room
- ⑥ Reverse U-metal Curved Ceiling
- ⑦ Comfort Seating & Living Areas
- ⑧ Thermal Comfort Flow Nodes
- ⑨ Spiral Staircase from Main Level
- ⑩ Fiber Optic Air and Light Delivery Node (integrates with the existing light tubes)

